

SPC vs Specification - Control Limits vs Spec Limits

Why mixing up control limits and spec limits is the #1 SPC mistake

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The Most Common SPC Mistake

- Setting control chart limits equal to USL and LSL - this destroys the entire purpose of SPC.
- Spec limits define what the customer needs. Control limits define what the process actually does.
- If $UCL=USL$ and $LCL=LSL$, the chart will NEVER show an out-of-control signal until the process already makes scrap.
- An IATF auditor will immediately identify this as a Major NC.

Control Limits - Process Voice

Definition:	Statistical limits calculated from actual process data. $UCL = \bar{X} + 3\sigma$. $LCL = \bar{X} - 3\sigma$.
Purpose:	Detect special causes. Signal when process behavior has changed. Action signal for operators.
Calculation:	From 25+ subgroups of real production data. $\sigma = \bar{R}/d_2$ for X-bar chart.
Width:	Control limits are narrow when the process is precise. Wide when process has high variation.
Rule:	NEVER set control limits from USL/LSL. ALWAYS calculate from process data.

Spec Limits - Customer Voice

Definition:	The engineering tolerance band: USL (Upper Spec Limit) and LSL (Lower Spec Limit).
Purpose:	Define conformance. Parts outside spec = non-conforming. Separate concept from control.
Source:	Engineering drawing, customer specification, design FMEA output.
On the Chart:	Draw as dashed lines for reference only. NEVER as control limits.
Cpk Link:	Cpk measures how far the process mean is from the nearest spec limit, in sigma units.

Capability Index Quick Reference

Cp:	$(USL - LSL)/(6\sigma_{short})$ - Measures spread. Assumes process is centered.
Cpk:	$\min[(USL - \bar{x})/3\sigma, (\bar{x} - LSL)/3\sigma]$ - Accounts for centering. Use for ongoing.
Pp:	$(USL - LSL)/(6\sigma_{long})$ - Uses STDEV of all readings. For initial study.
Ppk:	$\min[(USL - \bar{x})/3\sigma_{long}, (\bar{x} - LSL)/3\sigma_{long}]$ - PPAP requirement: $Ppk \geq 1.67$ for CC.
Cpk ≥ 1.67:	Required for CC characteristics. Process is 5-sigma capable.
Cpk 1.33-1.67:	Acceptable for SC characteristics. Process is 4-sigma capable.
Cpk < 1.33:	Unacceptable for any special characteristic. Immediate improvement required.